

EXPERIMENT 2- File Copy Using System Calls

nano exp2.c

code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;

    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        if (write(destination, buffer, bytesRead) != bytesRead)
        {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }

    close(source);
    close(destination);
    return 0;
}
```

```

    if (write(destination, buffer, bytesRead) != bytesRead)
    {
        perror("Error writing file");
        close(source);
        close(destination);
        return 1;
    }
}

if (bytesRead < 0)
{
    perror("Error reading file");
}

close(source);
close(destination);

printf("File copied successfully.\n");

return 0;
}

```

echo "Hello, this is my OSSP system call experiment." > source.txt

cat source.txt

output:

cat source.txt

File copied successfully.//output

```

bharath@DESKTOP-6D34C2R:~/OSSP$ cd ~/OSSP
nano exp2.c
bharath@DESKTOP-6D34C2R:~/OSSP$ echo "Hello, this is my OSSP system call experiment." > source.txt
bharath@DESKTOP-6D34C2R:~/OSSP$ cat source.txt
Hello, this is my OSSP system call experiment.
bharath@DESKTOP-6D34C2R:~/OSSP$ gcc exp2.c -o exp2
bharath@DESKTOP-6D34C2R:~/OSSP$ ./exp2
File copied successfully.
bharath@DESKTOP-6D34C2R:~/OSSP$ cat destination.txt
Hello, this is my OSSP system call experiment.
bharath@DESKTOP-6D34C2R:~/OSSP$ cat sample.txt

```

What are we doing?

We create a C program that copies one file to another.

Example:

source.txt



destination.txt

We use four system calls:

open()

read()

write()

close()

open()

Opens the file.

open → Open file

read()

Reads the contents.

File → read → Program

write()

Writes the contents into another file.

Program → write → New file

close()

Closes the file.

close → File closed

Why do we do this?

To understand how a program communicates with the Linux operating system to access files.

Your practical also asks you to use strace to see the system calls made by cat sample.txt.

OSSP Practical Experiments

What is strace?

strace shows the system calls happening behind a command.

For example:

Bash

strace `cat` sample.txt

You may see:

openat()

read()

write()

close()

Remember:

open = open

read = read data

write = write data

close = close